Cloud Computing Lab1

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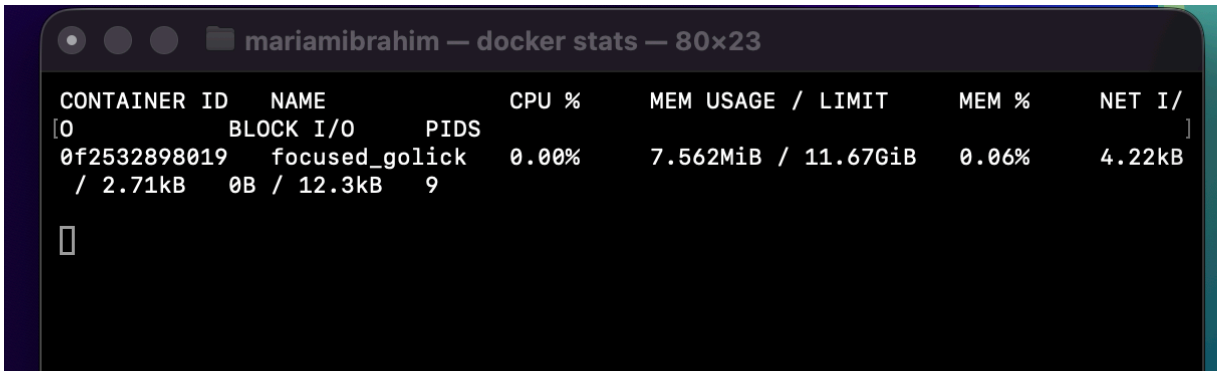
SE2

Deliverable 1:

Docker stats output (7.56 MiB, 9 processes)



Multipass VM outpu

- For the VM : use
  better and easier

Launched a new EC2 instance on AWS observed that Nitro was visible but not accessible
Ran the following command: sudo dnf install -y htop
To use htop to view processes

```
   ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
    ~~       V~' '->
     ~~~         /
       ~~._.   _/
          _/ _/
        _/m/'
[ec2-user@ip-172-31-20-197 ~]$ systemd-detect-virt
amazon
[ec2-user@ip-172-31-20-197 ~]$ lspci
ls /dev/nvme*
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma]
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.3 Non-VGA unclassified device: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:03.0 VGA compatible controller: Amazon.com, Inc. Device 1111
00:04.0 Non-Volatile memory controller: Amazon.com, Inc. NVMe EBS Controller
00:05.0 Ethernet controller: Amazon.com, Inc. Elastic Network Adapter (ENA)
/dev/nvme0  /dev/nvme0n1  /dev/nvme0n1p1  /dev/nvme0n1p127  /dev/nvme0n1p128
[ec2-user@ip-172-31-20-197 ~]$ sudo dmidecode -t bios
cat /proc/xen/xenbus 2>/dev/null
sudo: dmidecode: command not found
[ec2-user@ip-172-31-20-197 ~]$ sudo dnf install -y htop
htop
Amazon Linux 2023 Kernel Livepatch repository   268 kB/s |  31 kB     00:00
Dependencies resolved.
================================================================================
 Package    Architecture Version                        Repository         Size
================================================================================
Installing:
 htop       x86_64       3.2.1-87.amzn2023.0.3          amazonlinux       183 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 183 k
Installed size: 432 k
Downloading Packages:
htop-3.2.1-87.amzn2023.0.3.x86_64.rpm           4.5 MB/s | 183 kB     00:00
--------------------------------------------------------------------------------
Total                                           2.3 MB/s | 183 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : htop-3.2.1-87.amzn2023.0.3.x86_64                      1/1
  Running scriptlet: htop-3.2.1-87.amzn2023.0.3.x86_64                      1/1
  Verifying        : htop-3.2.1-87.amzn2023.0.3.x86_64                      1/1

Installed:
  htop-3.2.1-87.amzn2023.0.3.x86_64

Complete!
[ec2-user@ip-172-31-20-197 ~]$
```

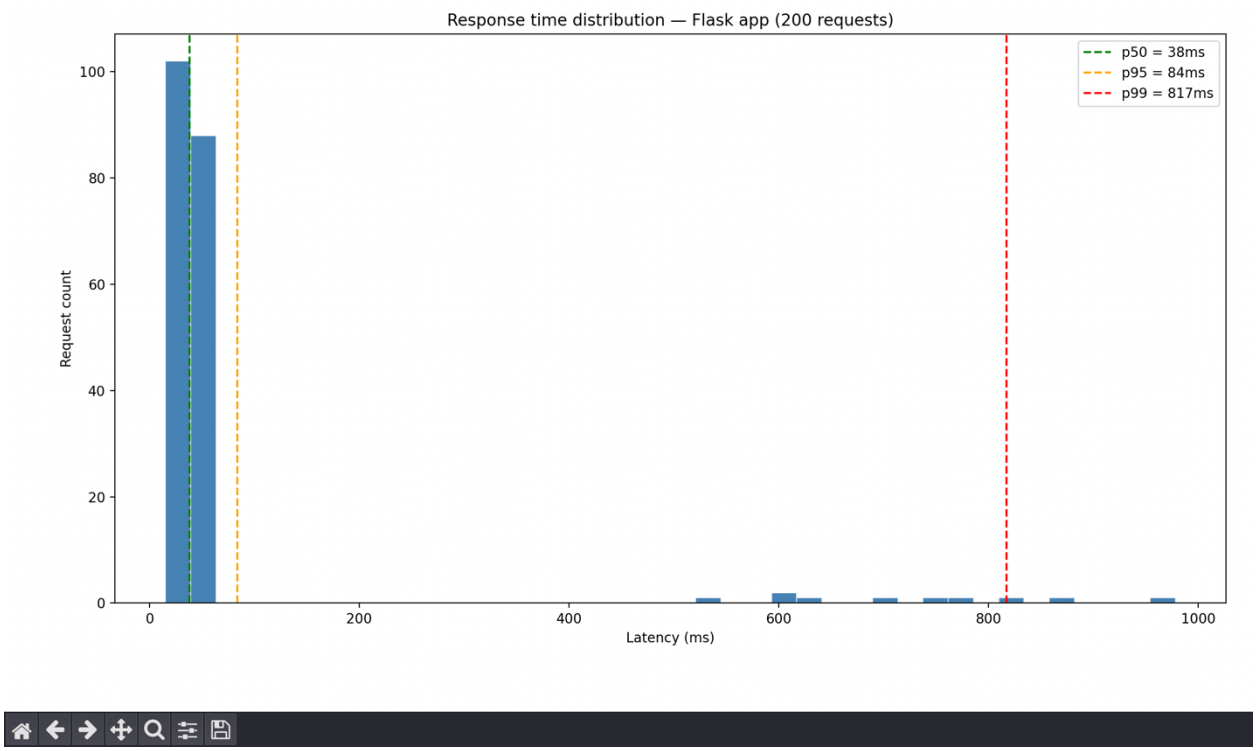
```
  0[                                                          0.0%] Tasks: 31, 14 thr, 71 kthr; 1 running
  1[|                                                         0.7%] Load average: 0.20 0.15 0.05
Mem[||||||||||||||||||||||||||||||||||||||||||||||||      147M/917M] Uptime: 00:07:15
Swp[                                                          0K/0K]

 Main   I/O
   PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
     1 root       20   0  169M 17808 11096 S  0.0  1.9  0:00.91 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
   838 root       20   0 54276 15176 13944 S  0.0  1.6  0:00.16 /usr/lib/systemd/systemd-journald
  1235 root       20   0 32604 11432  8592 S  0.0  1.2  0:00.07 /usr/lib/systemd/systemd-udevd
  1253 systemd-re 20   0 23508 16020 12208 S  0.0  1.7  0:00.08 /usr/lib/systemd/systemd-resolved
  1256 root       16  -4 21920  2332  1464 S  0.0  0.2  0:00.01 /sbin/auditd
  1257 root       16  -4 21920  2332  1464 S  0.0  0.2  0:00.00 /sbin/auditd
  1390 root       20   0 17100  6656  5660 S  0.0  0.7  0:00.00 /usr/bin/systemd-inhibit --what=handle-suspend-key:handle-hibernate-key --who=noah --why=acpid instead --mode
  1393 root       20   0 81428  3152  2936 S  0.0  0.3  0:00.02 /usr/sbin/irqbalance --foreground
  1394 libstorage 20   0  2780  1992  1828 S  0.0  0.2  0:00.01 /usr/bin/lsmd -d
  1397 root       20   0 17604  8260  7128 S  0.0  0.9  0:00.02 /usr/lib/systemd/systemd-homed
  1398 root       20   0 19452 10276  7924 S  0.0  1.1  0:00.06 /usr/lib/systemd/systemd-logind
  1400 root       20   0 81428  3152  2936 S  0.0  0.3  0:00.00 /usr/sbin/irqbalance --foreground
  1420 dbus       20   0  8492  4136  3532 S  0.0  0.4  0:00.02 /usr/bin/dbus-broker-launch --scope system --audit
  1421 systemd-ne 20   0  232M 10364  8928 S  0.0  1.1  0:00.05 /usr/lib/systemd/systemd-networkd
  1423 dbus       20   0  5388  2916  2464 S  0.0  0.3  0:00.02 dbus-broker --log 4 --controller 9 --machine-id ec2de537ecd40dbb2d0cd5c56a2ddc95 --max-bytes 536870912 --max-
  1424 root       20   0  2692  1136  1048 S  0.0  0.1  0:00.00 /usr/sbin/acpid -f
  1428 root       20   0  276M  3844  2812 S  0.0  0.4  0:00.00 /usr/sbin/gssproxy -D
  1429 root       20   0  276M  3844  2812 S  0.0  0.4  0:00.00 /usr/sbin/gssproxy -D
  1430 root       20   0  276M  3844  2812 S  0.0  0.4  0:00.00 /usr/sbin/gssproxy -D
  1431 root       20   0  276M  3844  2812 S  0.0  0.4  0:00.00 /usr/sbin/gssproxy -D
  1432 root       20   0  276M  3844  2812 S  0.0  0.4  0:00.00 /usr/sbin/gssproxy -D
  1433 root       20   0  276M  3844  2812 S  0.0  0.4  0:00.00 /usr/sbin/gssproxy -D
  1556 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.16 /usr/bin/amazon-ssm-agent
  1560 root       20   0 15260  9400  7988 S  0.0  1.0  0:00.00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
  1567 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.01 /usr/bin/amazon-ssm-agent
  1568 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.00 /usr/bin/amazon-ssm-agent
  1569 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.04 /usr/bin/amazon-ssm-agent
  1580 root       20   0  4784  2632  2444 S  0.0  0.3  0:00.00 /usr/sbin/atd -f
  1581 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.05 /usr/bin/amazon-ssm-agent
  1582 root       20   0  216M  1088  1000 S  0.0  0.1  0:00.00 /sbin/agetty -o -p -- \u --noclear - linux
  1583 root       20   0  216M  1092  1000 S  0.0  0.1  0:00.00 /sbin/agetty -o -p -- \u --keep-baud 115200,57600,38400,9600 - vt220
  1587 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.00 /usr/bin/amazon-ssm-agent
  1590 chrony     20   0 86340  3236  2668 S  0.0  0.3  0:00.02 /usr/sbin/chronyd -F 2
  1597 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.00 /usr/bin/amazon-ssm-agent
  1689 root       20   0 1212M 18768 10640 S  0.0  2.0  0:00.00 /usr/bin/amazon-ssm-agent
  1740 root       20   0 16868 11200  9196 S  0.0  1.2  0:00.02 sshd: ec2-user [priv]
  1744 root       20   0 17100  6660  5664 S  0.0  0.7  0:00.03 /usr/lib/systemd/systemd-userdbd
  1761 root       20   0 17460  7124  6120 S  0.0  0.8  0:00.00 systemd-userwork: waiting...
  1762 root       20   0 17692  7840  6740 S  0.0  0.8  0:00.00 systemd-userwork: waiting...
  1763 root       20   0 17692  7924  6824 S  0.0  0.8  0:00.00 systemd-userwork: waiting...
  1796 ec2-user   20   0 22808 14252 10600 S  0.0  1.5  0:00.06 /usr/lib/systemd/systemd --user
  1798 ec2-user   20   0  170M  6768     0 S  0.0  0.7  0:00.00 (sd-pam)
  1805 ec2-user   20   0 16868  6916  4916 S  0.0  0.7  0:00.02 sshd: ec2-user@pts/0
  1806 ec2-user   20   0  218M  5088  3644 S  0.0  0.5  0:00.02 -bash
  2671 ec2-user   20   0  218M  4668  3268 R  0.7  0.5  0:00.26 htop

F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

Deliverable 2: Latency Histogram

```
mariamibrahim@mariams-air ~ %
python3 load_test.py Progress: 0/200
Progress: 20/200 Progress: 40/200
Progress: 60/200 Progress: 80/200
Progress: 100/200 Progress: 120/200
Progress: 140/200 Progress: 160/200
Progress: 180/200
p50=38.4ms p95=531.4ms
p99=871.3ms Max=978.1ms
Mean=72.0ms
```



Deliverable 3:

Containers are better for microservices because of their low memory footprint, near-instant startup, and high density. A server that runs 1–2 VMs can run hundreds of containers. However, VMs are preferred when strong isolation, different kernels, or compliance requirements exist.

The VM (Multipass/Ubuntu) used 335 MB of memory and ran 109 processes just to exist. The container (Docker) used only 7.56 MiB and ran 9 processes — a 44x difference in memory and 12x in process count. VM startup took ~30 seconds vs under 1 second for the container.

Lab 2